Network Analysis
BEG225EL

Year: II **Semester: III**

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal Assessment		Final		Total
			Theory	Practical	Theory	Practical	
3	1	2	20	25	80	-	125

Course Objective: To provide the basis for formula and solution of network equations and to develop one-port and two port networks with given network functions.

Course Contents:

1. **Network Analysis** **(2 hours)**
    1.1. Review of network: Mesh and Nodal-pair

2. **Circuit Dynamics** **(8 hours)**
    2.1. First order RL and RC circuits
    2.2. Complete response of RL and RC circuit to sinusoidal input
    2.3. RLC circuit: Step response of RLC circuits, Response of RLC circuit to sinusoidal inputs, Resonance, Damping factors and Q-factor.

3. **Laplace Transform and Electrical Network Solutions** **(6 hours)**
    3.1. Definition and properties of Laplace transform of common forcing functions
    3.2. Initial and final value theorem
    3.3. Inverse Laplace transform: Partial fraction expansion
    3.4. Solutions of first order and second order system, RL and RC circuit, RLC circuit
    3.5. Transient and steady-state responses of network to: unit step, unit impulse, ramp and sinusoidal forcing functions

4. **Transfer Functions** **(6 hours)**
    4.1. Transfer functions of network system
    4.2. Poles and Zeros plot and analysis
    4.3. Time-domain behavior from pole-zero locations
    4.4. Stability and Routh's Criteria, Network stability

5. **Fourier Series and Transform** **(4 hours)**
    5.1. Evaluation of Fourier coefficients for periodic sinusoidal and non-sinusoidal waveforms
    5.2. Fourier Transform: Application of Fourier transforms for non-periodic waveforms

6. **Frequency Response of System** **(4 hours)**
    6.1. Magnitude and phase spectrums
    6.2. Bode plots and its applications

7. **One-Port Passive Network** **(8 hours)**
    7.1. Properties of one-port passive network
    7.2. Driving point functions: Positive Real Function, lossless network synthesis of LC one-port network
    7.3. Properties of RL and RC network, Synthesis of RL and RC network
    7.4. Properties and synthesis of RLC one-port network

8. **Two-Port Passive Network** **(7 hours)**

   8.1. Properties of two-port network: Reciprocity and symmetry

   8.2. Short circuit and open circuit parameters, transmission parameters, Hybrid parameter

   8.3. Relation and transformations between sets of parameters, Synthesis of two-port LC and RC ladder network

Practicals:

1. Transient and steady state responses of first order Passive network
2. Transient and Steady state responses of second order Passive network
3. Measurement of Frequency responses of first order and second order circuits
4. Measurement of Harmonic content of a waveform
6. Synthesis of one-port network functions and verify the responses using oscilloscope

Reference Books:

1 M. E. Van Valkenburg, "*Network Analysis*", 3rd Edition, Prentice Hall of India, 1995

2. M. L. Soni & J. C. Gupta, "*A Course in Electrical Circuit Analysis*", Dhanapat Rai & Sons, India

3. K. C. Ng., "*Electrical Network Theory*", A. H. Wheeler & Company (P) Ltd., India